

ChE 417 – Data, Statistics, Modeling for Engineers and Scientists

Quiz 1 – (10 points)

Problem 1 – Teaching performance

5 points

A ChSE professor calculated final grades in his class and was unhappy with student performance. As a result, he implemented a new strategy, and he thinks it can lead to better grades. He would like to compare student performance between the previous and current semesters using the 90th percentile of the grades. Use data - **grades_data.dat** with 1st column = data from previous semester and 2nd column = data from current semester.

Questions –

1. Compare percentile curves for different % values for both semesters. Determine 90% percentile for both based on these curves. Do it yourself from data, do not use in-built functions.
2. Determine whether the new strategy helped lead to improvements in student performance or not?
3. Will the assessment change if different percentiles (%) are used for comparison?

Problem 2 – Clinical Trials

5 points

A pharma company is trying out a new drug to reduce high blood pressure. To test the drug, the company conducted a clinical trial with a small group of 10 people and computed the average pressure. The cost of the test for each person is \$200. To ensure the test is robust, they repeated the trial 5 times (with 5 different groups – **clinical_smallgroup.dat**). The average pressure across trials is often called the mean estimate, and the SD across trials is a measure of robustness or reproducibility (precision). A widely used metric of precision is the coefficient of variation (COV) (ratio of SD and mean estimate). The company deems the study is consistent if the COV is less than 1.

1. Determine the average pressure for each trial. How much variability do you see across trials?
2. Determine COV. Is the study robust? What is the overall cost of the study?
3. Repeat the procedure for larger groups of 100 people (and 5 trials) – **clinical_largegroup.dat**. Is this study robust? What is the cost? Is the cost justified?